

SECTION - I
Multiple Choice Questions (MCQ)

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- The coordinates of the vertices of a tetrahedron ABCD are as follows A (2, 3, 4); B (1, -1, 2); C (0, 4, 5); D(-2, 3, -4). Then, which of the following is true -
 (1) the angle between lines AB and CD is $\cos^{-1} \left(\frac{24}{\sqrt{1860}} \right)$
 (2) the angle between AD and the plane ABC is $\sin^{-1} \sqrt{\frac{80}{11}}$
 (3) the equation of line BD is $\frac{x-1}{3} = \frac{y-1}{-4} = \frac{z-2}{6}$
 (4) the perpendicular distance from D to the plane ABC is $\frac{80}{\sqrt{110}}$
- If $\vec{a}$, $\vec{b}$, $\vec{c}$ represents adjacent edges of tetrahedron such that $|\vec{a}| = |\vec{b}| = |\vec{c}| = 2$ and angle between $\vec{a}$ and $\vec{b}$ is $\pi/3$, and $\vec{c}$ is $\pi/3$ and $\vec{a}$ and $\vec{c}$ $\pi/3$ then its volume is
 (1) $4\sqrt{2}$ (2) $\sqrt{\frac{2}{3}}$ (3) $\frac{2\sqrt{2}}{\sqrt{3}}$ (4) $\frac{2\sqrt{2}}{3}$
- If $\vec{e}_1 = (1, 1, 1)$, $\vec{e}_2 = (1, 1, -1)$ and $\vec{a}$ and $\vec{b}$ are two vectors such that $\vec{e}_1 = 2\vec{a} + \vec{b}$ and $\vec{e}_2 = \vec{a} + 2\vec{b}$, then the angle between $\vec{a}$ & $\vec{b}$ is
 (1) $\cos^{-1} \left(\frac{7}{9} \right)$ (2) $\cos^{-1} \left(\frac{7}{11} \right)$ (3) $\cos^{-1} \left(-\frac{7}{11} \right)$ (4) $\sin^{-1} \left(\frac{6\sqrt{2}}{11} \right)$
- If $|\vec{a}| = 3$, $|\vec{b}| = 4$, $|\vec{c}| = 5$ and $\vec{a} \perp (\vec{b} + \vec{c})$, $\vec{b} \perp (\vec{c} + \vec{a})$, $\vec{c} \perp (\vec{a} + \vec{b})$ then $|\vec{a} + \vec{b} + \vec{c}|$ is equal to-
 (1) 50 (2) 25 (3) $5\sqrt{2}$ (4) 12
- The shortest distances of a point from x-axis, y-axis and z-axis are 2, 3, 6 respectively. The distance of this point from the origin is
 (1) $\frac{7}{\sqrt{2}}$ (2) 7 (3) 11 (4) $\frac{49}{2}$
- Let a line having direction ratios 1, -4, 2 intersect the lines $\frac{x-7}{3} = \frac{y-1}{-1} = \frac{z+2}{1}$ and $\frac{x}{2} = \frac{y-7}{3} = \frac{z}{1}$ at the point A and B. Then $(AB)^2$ is equal to _____.
 (1) 72 (2) 70 (3) 11 (4) 84

7. The perpendicular distance of a corner of a unit cube from the diagonal not passing through that corner, is
- (1) $\frac{2}{3}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\sqrt{\frac{2}{3}}$ (4) $\frac{1}{\sqrt{2}}$
8. If the shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{\lambda}$ and $\frac{x-2}{1} = \frac{y-4}{4} = \frac{z-5}{5}$ is $\frac{1}{\sqrt{3}}$, then the sum of all possible values of λ is :
- (1) 16 (2) 6 (3) 12 (4) 15
9. The shortest distance (S.D) between lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$ is
- (1) $3\sqrt{30}$ (2) $\sqrt{30}$
 (3) $2\sqrt{30}$ (4) $4\sqrt{30}$
10. The angle between lines whose D.C.'s satisfy the following equation $\ell + m + n = 0$, $\ell^2 + m^2 - n^2 = 0$ is
- (1) $\frac{\pi}{4}$ (2) $\frac{\pi}{6}$
 (3) $\frac{5\pi}{6}$ (4) $\frac{\pi}{3}$
11. If the two lines $\ell_1 : \frac{x-2}{3} = \frac{y+1}{-2}, z = 2$ and $\ell_2 : \frac{x-1}{1} = \frac{2y+3}{\alpha} = \frac{z+5}{2}$ perpendicular, then an angle between the lines ℓ_2 and $\ell_3 : \frac{1-x}{3} = \frac{2y-1}{-4} = \frac{z}{4}$ is :
- (1) $\cos^{-1}\left(\frac{29}{4}\right)$ (2) $\sec^{-1}\left(\frac{29}{4}\right)$ (3) $\cos^{-1}\left(\frac{2}{29}\right)$ (4) $\cos^{-1}\left(\frac{2}{\sqrt{29}}\right)$
12. If a line is such that it is inclined with y-axis and z-axis at 60° , then at what angle is it inclined with x-axis?
- (1) 45° (2) 30° (3) 75° (4) 60°
13. If the lines $x = ay + b$, $z = cy + d$ and $x = a'z + b'$, $y = c'z + d'$ are perpendicular, then
- (1) $ab' + bc' + 1 = 0$ (2) $cc' + a + a' = 0$ (3) $aa' + c + c' = 0$ (4) $bb' + cc' + 1 = 0$
14. The symmetric form of the line $3x + 2y - z - 4 = 0$, $4x + y - 2z + 3 = 0$ is
- (1) $\frac{x+2}{3} = \frac{y-5}{-2} = \frac{z}{5}$ (2) $\frac{x-2}{3} = \frac{y-5}{-2} = \frac{z}{5}$ (3) $\frac{x-2}{3} = \frac{y-5}{2} = \frac{z}{5}$ (4) $\frac{x+2}{3} = \frac{y-5}{2} = \frac{z-0}{-5}$
15. The length of the perpendicular from the point $(2, -1, 4)$ on the straight line, $\frac{x+3}{10} = \frac{y-2}{-7} = \frac{z}{1}$ is
- (1) Greater than 3 but less than 4 (2) Greater than 2 but less than 3
 (3) Greater than 4 (4) Less than 2
16. If ℓ_1, m_1, n_1 , and ℓ_2, m_2, n_2 are D. C's of the two lines inclined to each other at an angle θ , then the D.C's of the internal and external bisectors of the angle between these lines are
- (1) $\frac{\ell_1 + \ell_2}{2\sin(\theta)}, \frac{m_1 + m_2}{2\sin(\theta)}, \frac{n_1 + n_2}{2\sin(\theta)}$ (2) $\frac{\ell_1 + \ell_2}{\cos\left(\frac{\theta}{2}\right)}, \frac{m_1 + m_2}{\cos\left(\frac{\theta}{2}\right)}, \frac{n_1 + n_2}{\cos\left(\frac{\theta}{2}\right)}$
 (3) $\frac{\ell_1 - \ell_2}{2\sin\left(\frac{\theta}{2}\right)}, \frac{m_1 - m_2}{2\sin\left(\frac{\theta}{2}\right)}, \frac{n_1 - n_2}{2\sin\left(\frac{\theta}{2}\right)}$ (4) None of these

17. The equation of perpendicular from (3, -1, 11) to the line $\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ is
- (1) $\frac{x-3}{1} = \frac{y+1}{-6} = \frac{z-11}{4}$ (2) $\frac{x-3}{1} = \frac{y+1}{6} = \frac{z-11}{4}$
- (3) $\frac{x+3}{-1} = \frac{y+1}{-6} = \frac{z-11}{4}$ (4) None of these
18. The area of the parallelogram whose diagonals represent the vectors $3\hat{i} + \hat{j} - 2\hat{k}$ and $\hat{i} - 3\hat{j} + 4\hat{k}$ is
- (1) $10\sqrt{3}$ (2) $5\sqrt{3}$ (3) 8 (4) 4
19. If $|\vec{a}| = |\vec{b}| = 1$ and $\vec{a} \cdot \vec{b} = \frac{1}{2}$, then
- (1) $\left(\vec{a} - \frac{\vec{b}}{2}\right) \parallel \vec{b}$ (2) $\vec{a} \parallel \vec{b}$ (3) $\left(\vec{a} - \frac{\vec{b}}{2}\right) \perp \vec{b}$ (4) $\vec{a} \perp \vec{b}$
20. In a triangle PQR, S and T are points on QR and PR respectively, such that QS = 3SR and PT = 4TR. Let M be the point of intersection of PS and QT. Find the ratio $\frac{QM}{MT}$
- (1) 3.75 (2) 3 (3) 16/3 (4) 15/3

SECTION-II

(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. A line L is perpendicular to the lines having direction ratios (3, 2, 1) and (1, 2, 3). The sum of direction cosines of line L is
22. If a point R(4, y, z) lies on the line segment joining the points P(2, -3, 4) and Q(8, 0, 10), then the distance of R from the origin is $a \times \sqrt{b}$ find b - a
23. The sum of squares of direction cosines of all the four diagonals of a unit cube is
24. If the shortest distance between the lines $x + 2\lambda = 2y = -12z$, $x = y + 4\lambda = 6z - 12\lambda$ is $4\sqrt{2}$ units, then a value of λ^{10} is :
25. The projection of any line segment on axes are 3, 4, 5 respectively, then square of length of the line segment is

ANSWER KEY DPP-20

1.	(4)	2.	(4)	3.	(3)	4.	(3)	5.	(1)	6.	(4)	7.	(3)
8.	(1)	9.	(1)	10.	(4)	11.	(2)	12.	(1)	13.	(3)	14.	(1)
15.	(1)	16.	(3)	17.	(1)	18.	(3)	19.	(3)	20.	(1)	21.	(0)
22.	(12)	23.	(4)	24.	(32)	25.	(50)						